Practical 3 Quiz

Which is/are the most likely mutational process/es explaining the pattern observed between the positions 1,417 and 1,438 (numbers according to the top bar of JalView?

- [ ] a. Duplication and loss
- [x] b. Insertion and deletion
- [ ] c. Inversion
- [x] d. Point mutation

Practical 3 Quiz

By looking at the results of the protein domain analyses, which molecular function would you assign to this set of sequences?

- [ ] a. Protein binding
- [ ] b. Catalytic activity
- [x] c. Extracellular matrix
- [ ] d. Electron transport

Practical 3 Quiz

Look at the 'Domain' and 'Repeat' rows at InterProScan results. List all the differences that you find, if any, between the three queried sequences.

*Annotate the **ungapped** positions that you want to talk about. Max.: 500 characters.*

Of the three proteins, Homo sapiens and Bos mutus are the most alike, since both keep the collagen-repeat pattern and the C-terminal Fib_collagen_C/NC1 region near aa 1130–1430. The main change is that Bos misses the early repeat segments seen in Homo (32–75, 91–149, 190–243) and shows repeats later (440–494, 539–597, 965–1019, 1109–1167). Micropterus is the most different because it includes an extra VWC domain at 31–89 and a total of six repeat blocks.

Example of correct answer.

Practical 3 Quiz

By looking at the initial part of the alignment (alignment positions 70 to 163), there is a conserved region shared between some sequences.

- Vertader ✓
- Fals

Practical 3 Quiz

Based on the InterProScan results, can you identify which matches (domain, repeat...) correspond to this region?

Hint: to check the position of the aligned amino acid in the unaligned protein sequence look at the information displayed at the bottom left of the JalView software.

- a. Non cytoplasmatic domain
- b. None of them
- c. Collagen triple helix repeat ✓

Practical 3 Quiz

How many sites have been trimmed in relation to the original alignment?

Resposta: 674

Many of you didn't answer correctly… you just had to count (with a grep command or using JalView) the length of the original alignment and the length of the trimmed one and write the difference.

Practical 3 Quiz

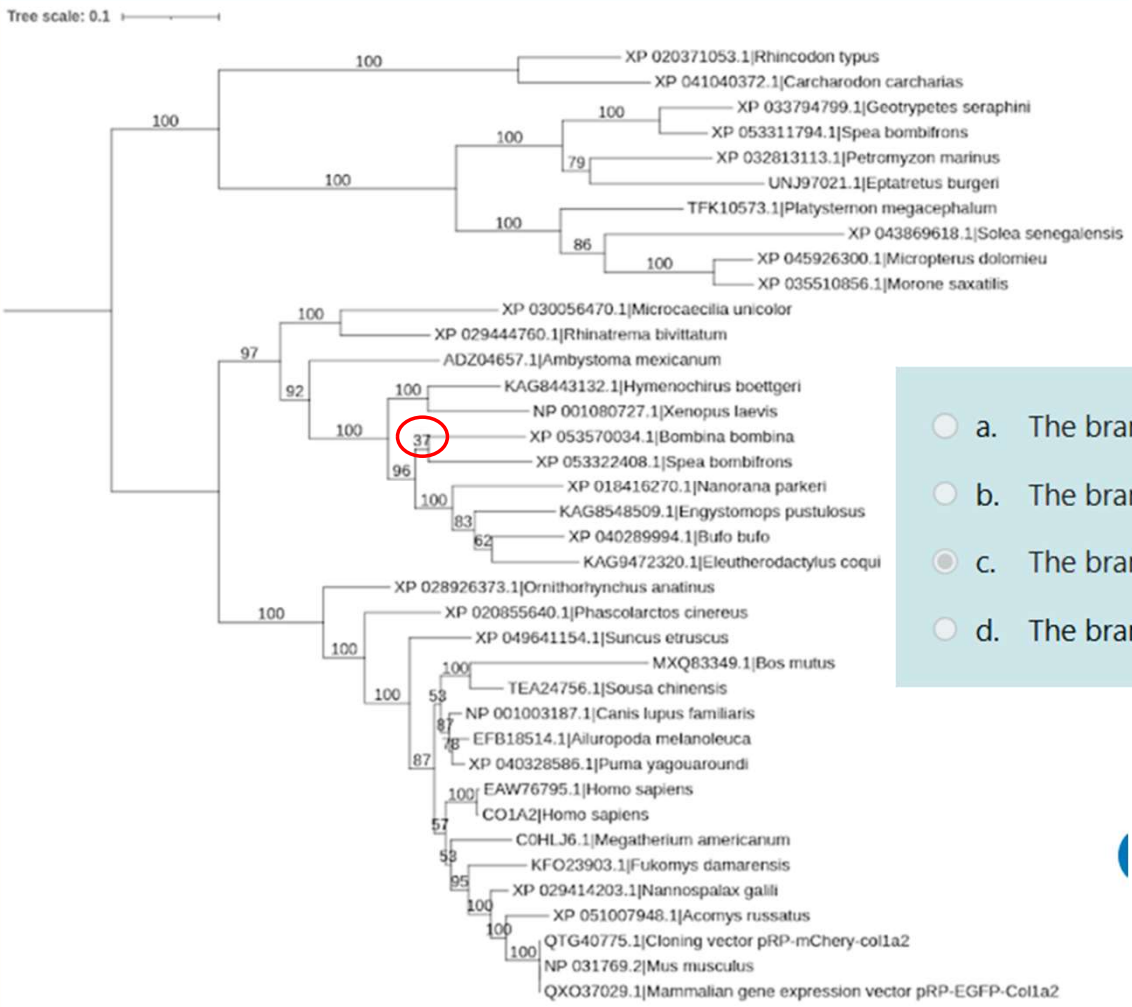
What could be the benefit from having trimmed a fraction of the alignment? Are there any potential drawbacks associated with this process?

- a. By trimming the alignment, we may have removed informative signal
- b. In general, we can expect alignment trimming to reduce computational time of the phylogenetic inference ✓
- c. Trimming the alignment will always improve the accuracy of the phylogenetic inference
- d. By trimming the alignment, there will be less informative sites, and this will always increase the computational time needed by the phylogenetic inference software to compute a phylogeny

Answer (a) might also be correct, but there was an error in the marking of the question, so every correct answer was considered as correct (only (a) = 1 point, only (b) = 1 point or both (a and b) = 1 point).

Practical 3 Quiz

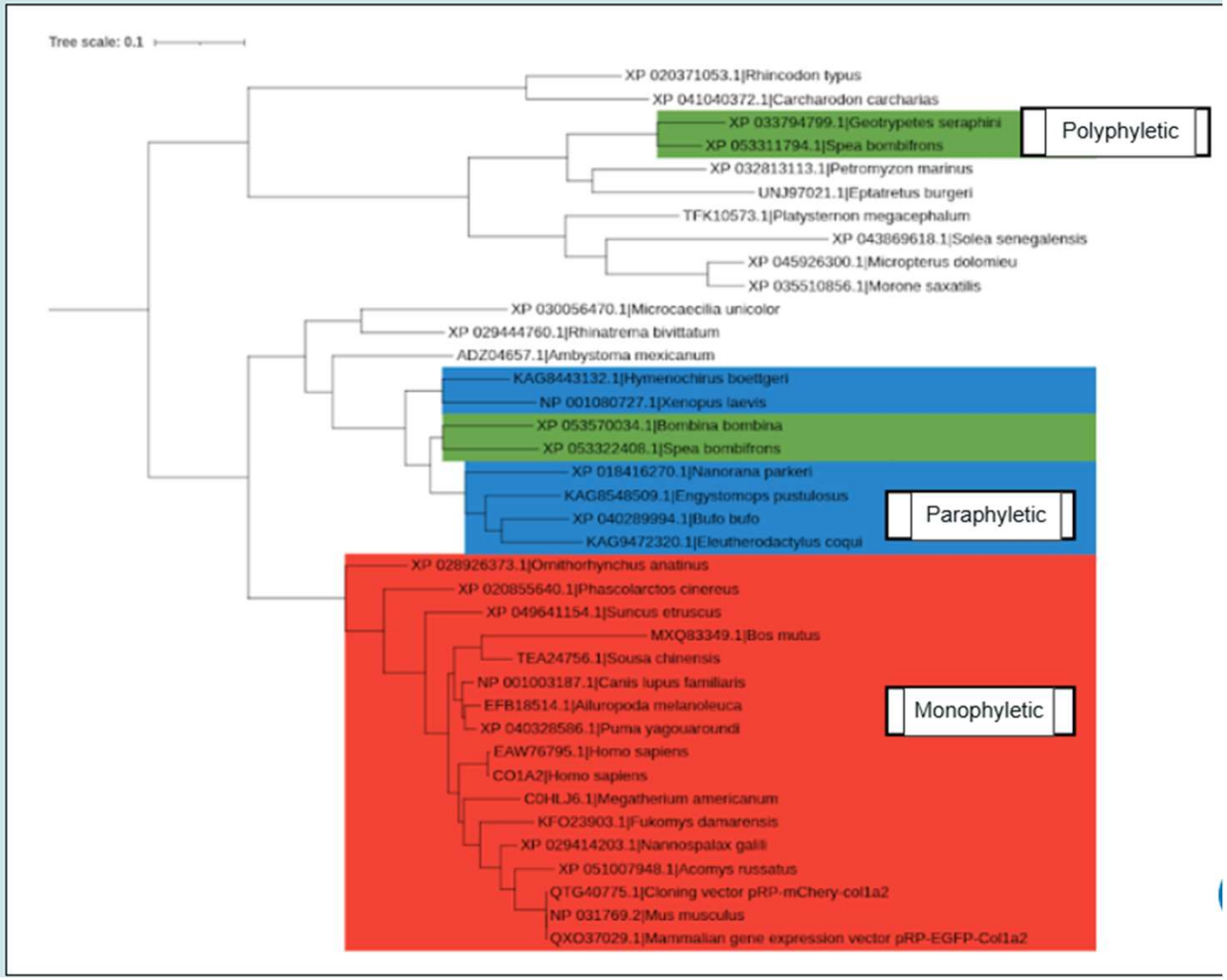
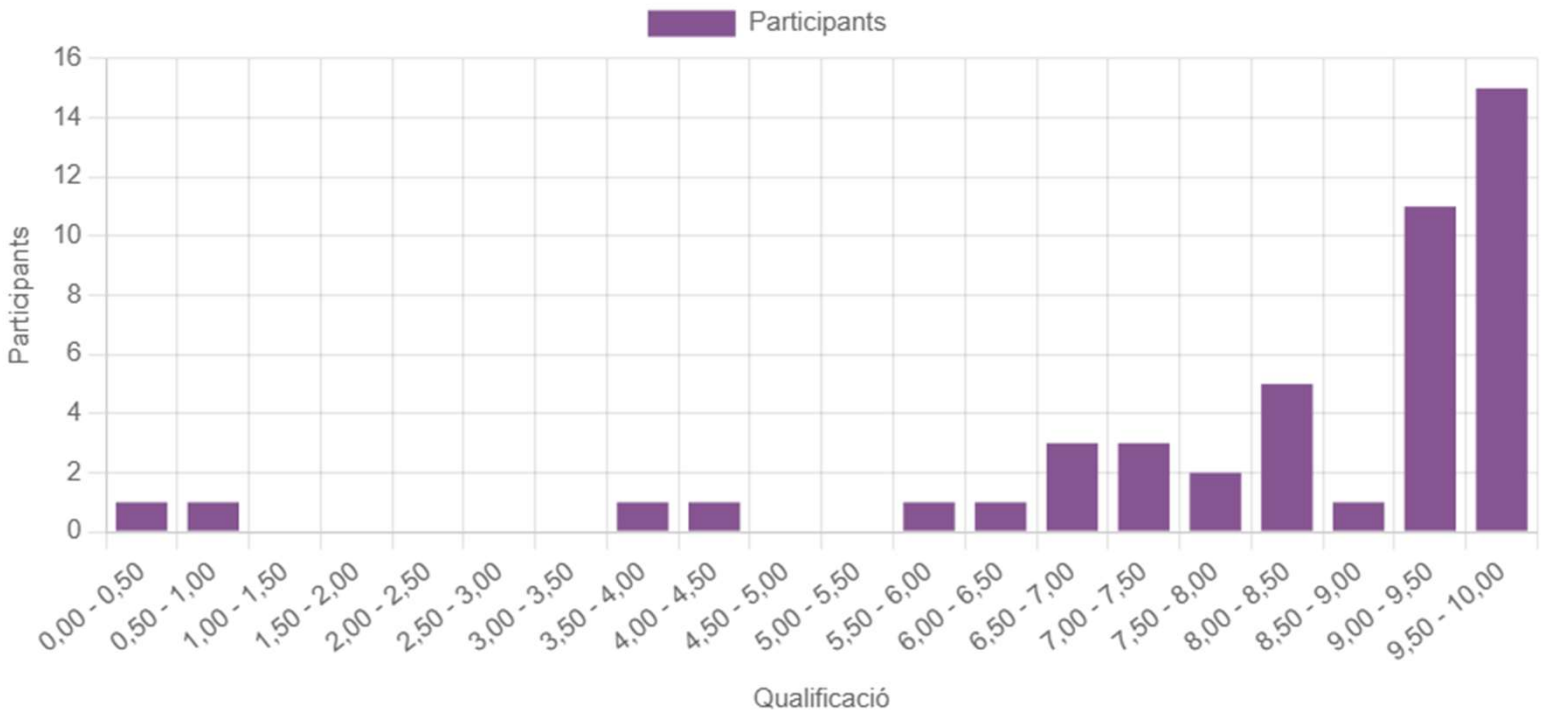
Which of the following branches would you trust the least?



- a. The branch separating *Megatherium americanum* and its sister group.
- b. The branch separating *Rhinatrema bivittatum* and *Microcaecilia unicolor*
- c. The branch separating *Spea bombifrons* and *Bombina bombina* ✓
- d. The branch separating the two human sequences.

Practical 3 Quiz

Match the corresponding coloured groups (blue, green and red) to their definition:



Practical 3 Quiz